Cboe®

cboe.com

Settlement of Standard, A.M.-Settled S&P 500 Index Options

Cboe Options Exchange (Cboe®) is the exclusive home for S&P 500 Index (SPX®) options trading, and SPX options are available in a variety of contract sizes, settlements and expirations. This piece sets forth information solely concerning settlement of standard (third Friday) SPX options that are A.M.-settled.

How is the Intraday S&P 500® Index Calculated?

S&P 500 Index values are calculated pursuant to the S&P's Index Mathematics Methodology and are based on the last trade price for each constituent stock. These intraday or "spot" values are calculated and disseminated on a real time basis throughout the trading day. The dissemination of intraday S&P Index values typically begins at 8:30 a.m. (CT).

The constituent stocks of the S&P 500 Index do not all open at the same time and each individual stock may not open for trading immediately at the open. As a result, the last trade price *from the prior trading day* is used for that constituent stock until it begins trading for a given day. Therefore, there are times (especially during the beginning of a trading session), when the intraday S&P 500 Index calculation is an amalgam of current trade prices and end-of-day prices from the prior trading day. When all constituent stocks have opened, the intraday S&P 500 Index is calculated using their current trade prices.

How is the Exercise Settlement Value for A.M.-Settled SPX Options Determined?

The exercise settlement value for A.M.-settled SPX options is based on a special opening quotation (SOQ) of the S&P 500 Index that is calculated on their expiration days (typically the third Friday of the expiration month).[1] The SOQ is based on the opening trade price in the primary market of each constituent stock in the S&P 500 Index on a given expiration (third) Friday.[2] The exercise-settlement amount for SPX options is equal to the difference between the exercise-settlement value (SET) and the exercise price of the option, multiplied by $100.

When is the SOQ Calculated on Expiration Days for SPX Options?

The SOQ is not calculated until all of the constituent stocks of the S&P 500 Index have opened for trading and their corresponding official opening prices are established. Because official opening prices are rarely disseminated at the opening, the SOQ (by construction) is not anchored to a specific time of day. As a result, the SOQ is typically calculated *after* intraday S&P 500 Index calculations have started and these intraday S&P 500 Index calculations typically

[1] Cboe publishes the exercise settlement value for A.M.-settled SPX options under the symbol "SET." For purposes of this piece, "SOQ" and "SET" are used interchangeable.
[2] If a constituent stock of the S&P 500 Index does not open on an SPX option expiration day, its last trade price from the prior trading day will be used in the SOQ.

include some stock prices from the prior trading day (especially during the beginning of a trading session).

Similarly, the SOQ is not associated with any particular intraday S&P Index value. Also, when the SOQ is calculated, the current values of the intraday S&P 500 Index and SOQ will typically be different. This is because the SOQ reflects opening trade prices, which may differ from the current trade prices of the constituent stocks of the S&P 500 Index at the time the SOQ is calculated.

Where Are the Official Opening Prices Established and How?

The SOQ is calculated pursuant to the S&P's Index Mathematics Methodology, which uses the trading day's official opening trade prices of the constituent stocks in the S&P 500 Index as set on the corresponding primary listing exchange. As of the date of this piece, the primary listing exchanges for the constituent S&P 500 Index stocks are NYSE, NASDAQ and Cboe.

NYSE, NASDAQ and Cboe conduct auctions at the open to start trading. Procedural details vary from one exchange to another; however, the goal of an opening auction is to open trading in a stock at the price that maximizes the quantity of shares transacted between buy and sell orders submitted for participation in the auction.

Why Are There Differences Between Intraday S&P 500 Index Values and the Exercise Settlement Value for SPX Options?

The prices of the constituent stocks of the S&P Index used in these calculations are different and that difference can yield different values for the calculations. As noted above, the SOQ uses official opening trade prices and it cannot be calculated until all the constituent stocks of the S&P 500 Index have opened on expiration days for SPX options.[3] In contrast, the calculation of intraday values of the S&P 500 Index begins at the open and starts as an amalgam of current trade prices and end-of-day prices from the prior trading day. When all constituent stocks have opened, the intraday S&P 500 Index is calculated using their current trade prices.

These distinctions set the SOQ apart from more familiar daily index milestones, such as the open, high, low or closing S&P 500 Index values, all of which are continuous calculations. A natural and known consequence of these differences is that the SOQ may fall outside of the open, high, low or closing values of the S&P 500 Index on SPX option expiration days. For example, if a significant market event is in process, the SOQ may be higher than the S&P 500 Index high for the trading session or lower than the S&P 500 Index low for the trading session.

What are Some Observed SOQ Behaviors?

Historically the deviation between the SOQ and the opening value of the S&P 500 Index has been less than 1%. By way of illustration, for the 61 March quarterly expiration days for A.M.-settled SPX options between March 2009 and March 2024, the distribution of such differences is centered at 0.09 percent, and it ranges from -- 0.98 percent to +1.00 percent. As described above, the SOQ can be above the high and below the low of the S&P 500 Index. During the 15-year sample period (March 2009 to March 2024), SOQ values fell outside the high and low range

[3] If a constituent stock of the S&P 500 Index does not open on an SPX option expiration day, its last trade price from the prior trading day will be used in the SOQ.

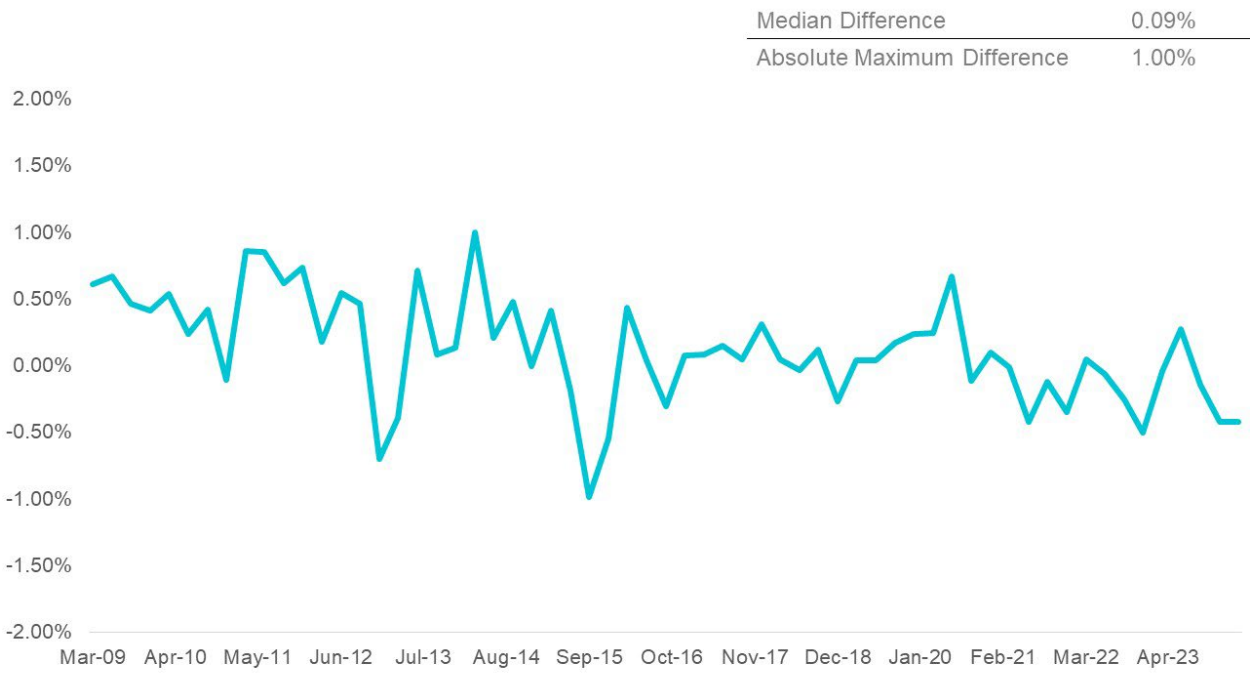
of the S&P 500 Index on quarterly A.M.-settled SPX option expiration dates approximately 30% of the time, most often above the daily high.

Historically, the deviations between the SOQ and: (1) the previous day's closing value of the S&P 500 Index, and (2) the closing value of the S&P 500 Index on SPX option expiration dates, appear relatively small. Between March 2009 and March 2024, the median difference between the SOQ and the previous day's S&P 500 Index close was 0.16%. Over the same period, the median difference between the SOQ and the S&P 500 Index close on SPX option expiration dates was 0.13%.

These data underscore the reality that neither the SOQ nor the opening and other intraday S&P 500 Index values (calculated before the SOQ is determined) are true "real-time values." Unless all constituent stocks of the S&P 500 Index trade simultaneously at the market open, the SOQ will use some constituent opening trade prices that are no longer current when the SOQ is calculated. Similarly, until all of the constituent stocks of the S&P 500 Index have opened for trading, the intraday calculation of the S&P 500 Index is an amalgam of current trade prices and end-of-day prices from the prior trading day. When all constituent stocks have opened, the intraday S&P 500 Index is calculated using their current trade prices.

Exhibit 1: SOQ Percentage Deviation from S&P 500 Index Open[4]

(SOQ minus S&P 500 Index Open) / (S&P 500 Index Open) March Quarterly SPX Option Expiration Dates, March 2009 through March 2024)



[4] Chart source: Cboe Hanweck.

Exhibit 2: SOQ vs. Current Day's S&P 500 Index High and S&P 500 Index Low[4]

For each March Quarterly SPX Option Expiration Date from March 2009 through March 2024, the SOQ (SET), S&P 500 Index High and Low[5] are normalized as percentages above or below the geometric mean of that day's S&P 500 Index Open and Close

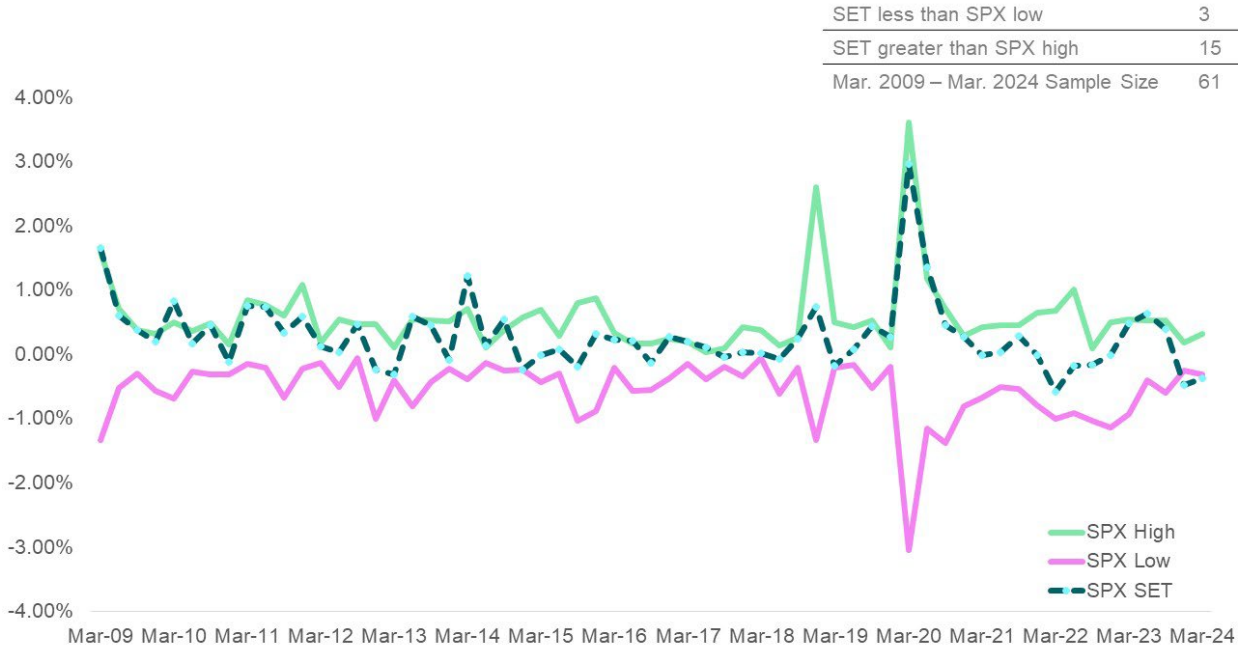
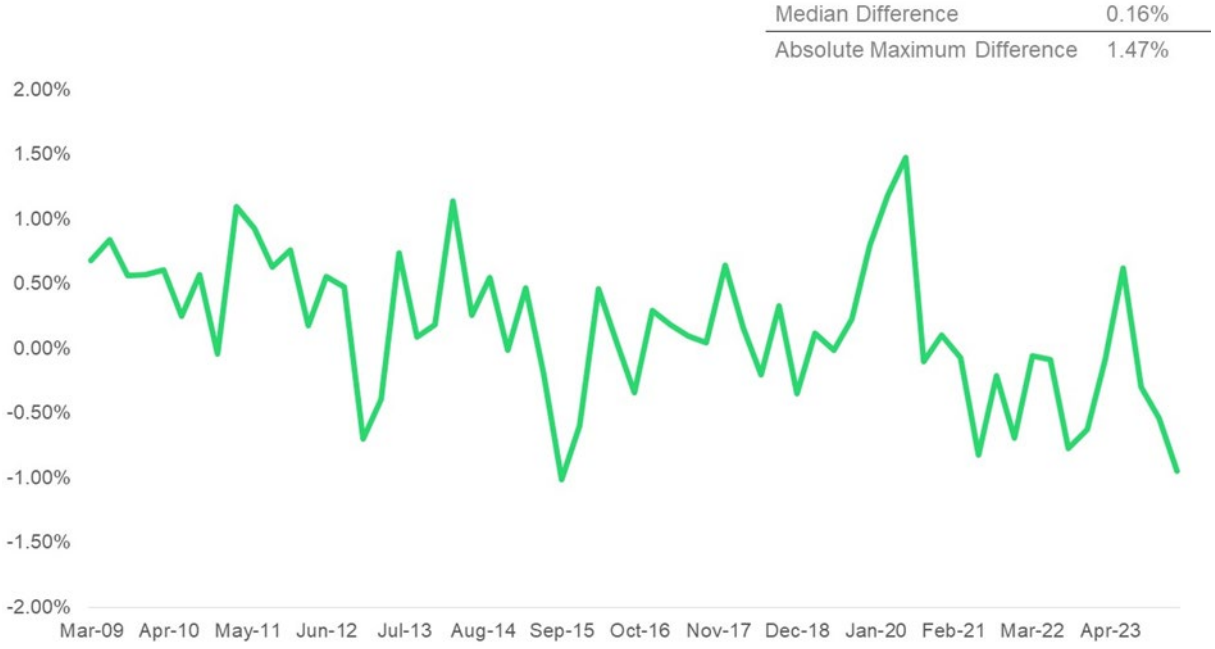


Exhibit 3: SOQ Deviation from Previous Day's S&P 500 Index Close[4]

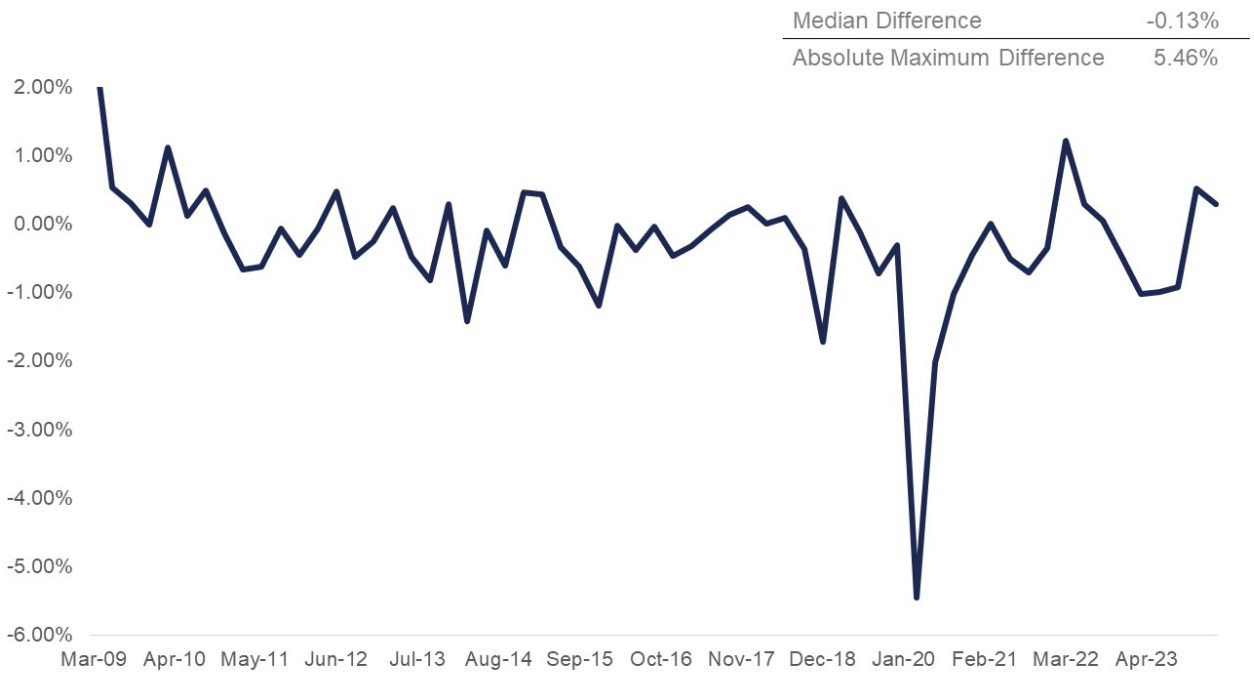
(SOQ minus Previous Day's S&P 500 Index Close)/(Previous Day's S&P 500 Index Close) - March Quarterly SPX Option Expiration Dates, March 2009 through March 2024)



[5] For purposes of Exhibit 2 only, "SPX" is used to mean "S&P 500 Index."

Exhibit 4: SOQ Deviation from Current Day's S&P 500 Index Close[4]

(SOQ Minus Current Day's S&P 500 Index Close) / (Current Day's S&P 500 Index Close) - March Quarterly SPX Option Expiration Days, March 2009 through March 2024)



The SOQ Enables a Tradable Settlement for SPX Options

On the days that A.M.-settled SPX options expire, S&P calculates an SOQ of the S&P 500 Index using the opening trade prices of the constituent stocks in their primary markets. Traders can replicate the exposure of their expiring SPX options by entering orders to buy and sell the constituent stocks of the S&P 500 Index at their opening trade prices. If they are successful, traders can effectively construct a portfolio that matches the value of the SOQ. At this point, the derivatives and cash markets converge.

Why Isn't the Opening S&P 500 Index Value Used for Settlement?

The SOQ is internally coherent and actionable in ways that the opening S&P 500 Index value is not. As described above, the use of an SOQ to settle SPX options allows for convergence of the derivatives and cash markets because the SOQ is calculated using actual opening trade prices. Conversely, as described above, because not all stocks have an opening trade price at the time the opening S&P 500 Index value is calculated, that value is not actionable. This is because the opening S&P 500 Index value is typically an amalgam of current trade prices and end-of-day prices from the prior trading day and is unrepresentative of actionable market valuations, especially on days when stock prices have gapped higher or lower at the open.

Because the SOQ is calculated using the official opening trade prices of the constituent stocks of the S&P 500 Index,[6] it is a comparatively direct reflection of the balance of actionable orders submitted on the primary listing exchanges for participation in their opening auctions.

Why Don't Standard SPX Options Settle to a Closing Value?

When cash-settled index options were first introduced in the 1980s, they generally utilized closing-price settlement procedures (i.e., P.M.-settlement). The U.S. Securities and Exchange Commission, however, had concerns about the impact of P.M.-settled, cash-settled index options on the underlying cash equities markets (e.g., market volatility and sharp price movements near the close on SPX option expiration days). As a result, beginning in 1987, a shift from P.M.-settlement (closing index values) to A.M.-settlement (using an SOQ) began,[7] which culminated in Cboe transitioning all standard (third Friday) SPX options to A.M. settlement in 1992.[8]

A Note on Quarterly (or "Triple Witching") Expirations

"Triple Witching" refers to the simultaneous expiration of individual stock options, equity index options and equity index futures that occurs four times a year on third Fridays in March, June, September and December. On these quarterly expiration days, there is often an observed increase in trading volume and market volatility as traders close, roll out or offset their expiring positions.

Cboe® and Cboe Global Markets® are registered trademarks of Cboe Exchange, Inc. or its affiliates. S&P®, SPX®, and S&P 500® are registered trademarks of Standard & Poor's Financial Services, LLC, and have been licensed for use by Cboe Exchange, Inc. All other trademarks and service marks are the property of their respective owners.

[6] If a constituent stock of the S&P 500 Index does not open on an SPX option expiration day, its last trade price from the prior trading day will be used in the SOQ.
[7] See Securities Exchange Act Release No. 24367 (April 17, 1987), 52 FR 13890 (April 27, 1987) (SR-CBOE-87-11).
[8] See Securities Exchange Act Release No. 30944 (July 21, 1992), 57 FR 33376 (July 28, 1992) (SR-CBOE-92-09).